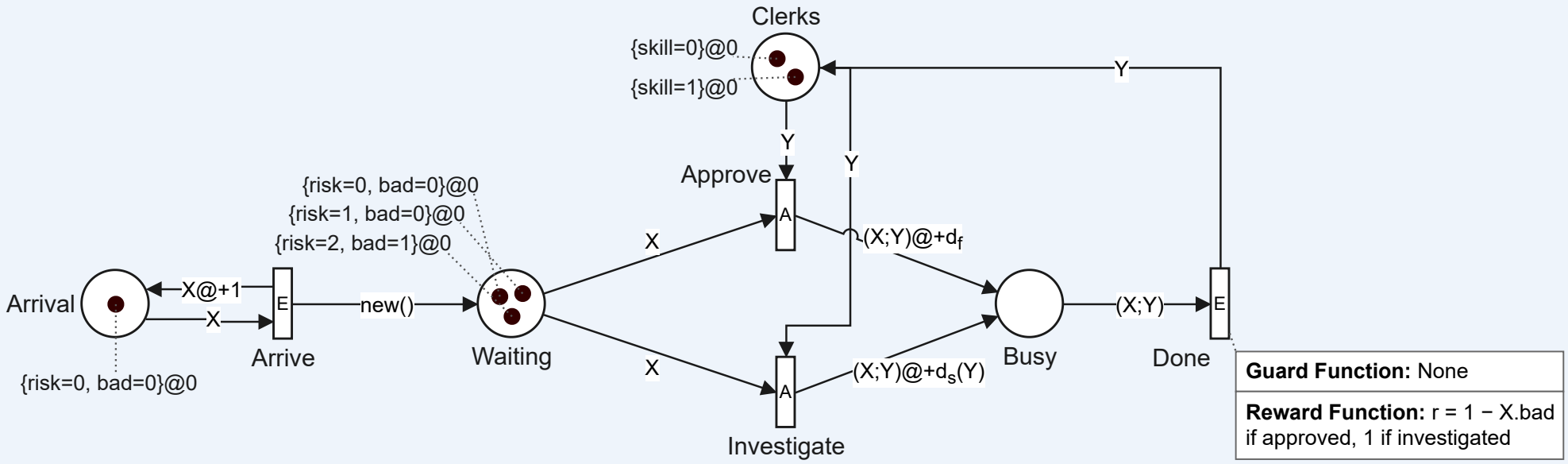
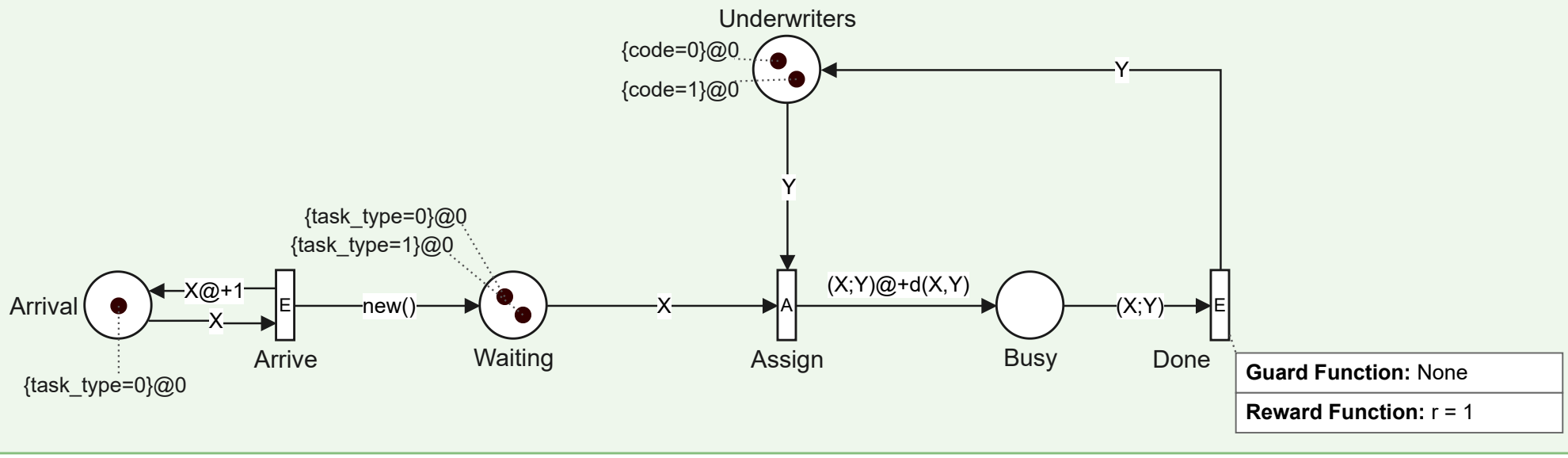
$(E, \tau = 0, \rho = 0)$

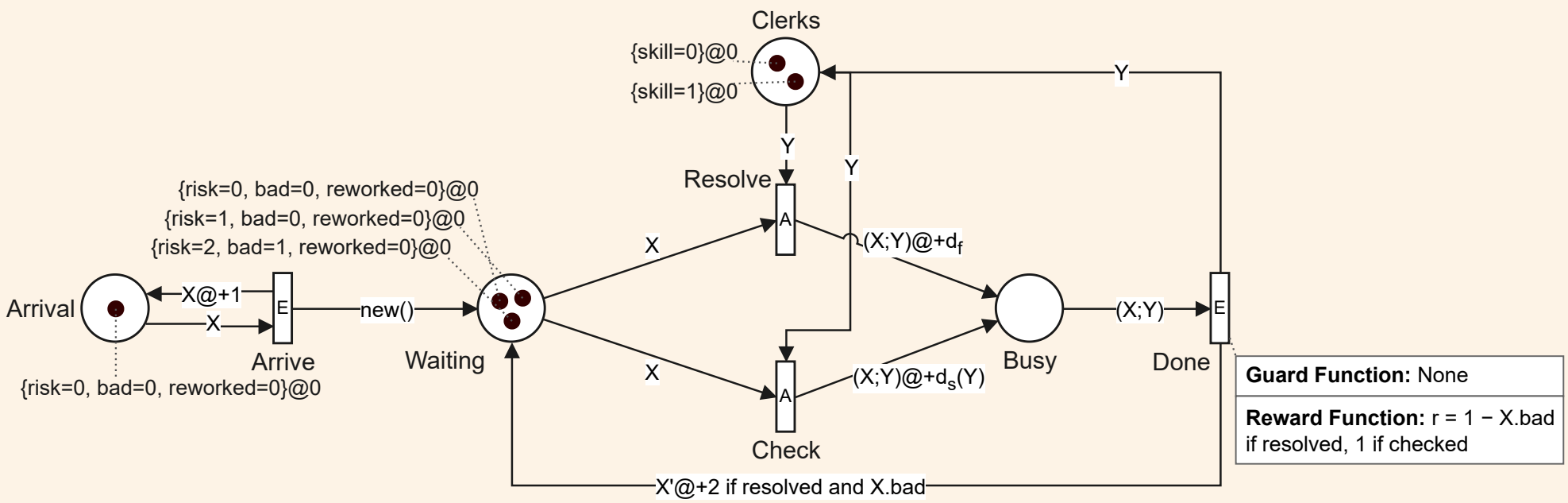
Component 1: Claims



Component 2: Underwriting



Component 3: Complaints



Definitions. A transition without a box has guard None and reward r = 0.
new(): a fresh case. Claims and complaints: risk ~ U{0,1,2}, bad ~ Bernoulli(p_{risk}) with p = (0.1, 0.5, 0.9), and reworked = 0 for complaints; bad is not observable.
Underwriting: task_type ~ U{0,1}.
d_f = 1 + U{0,1} (approve, resolve); $d_s(Y)$ = 4 + U{0,1} for a normal clerk (skill = 0) and 2 + U{0,1} for an expert (skill = 1) (investigate, check).
d(X,Y) = 1 + U{0,1} if X.task_type = Y.code, else 3 + U{0,1}.
X' = X with reworked = 1 and bad = 0: a wrongly resolved complaint reopens after 2 time units.